

Jens Athens E. Gumawid

Problem. A new fertilizer has been developed to increase the yield on crops, and the makers of the fertilizer want to better understand which of the three formulations (blends) of this fertilizer are most effective for wheat, corn, soybeans and rice (crops). They test each of the three blends on 5 samples of each of the four types of crops. The crop yields for the 12 combinations are as shown in the table below.

Crop				
Fertilizer	Wheat	Corn	Soy	Rice
Blend X	123	128	166	151
	156	150	178	125
	112	174	187	117
	100	116	153	155
	168	109	195	158
Blend Y	135	175	140	167
	130	132	145	183
	176	120	159	142
	120	187	131	167
	155	184	126	168
Blend Z	156	186	185	175
	180	138	206	173
	147	178	188	154
	146	176	165	191

	193	190	188	169
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Fertilizer and Crop Yield: Two-Way ANOVA

The study shows how different fertilizer blends affect the yield of four crops including wheat, corn, soy, and rice. The purpose is to find out whether the type of fertilizer used has an effect on each crop yield, and its effects depending on the crop. In this case, a crop yield is the dependent variable, while the fertilizer blend (X, Y, and Z) and crop type serve as the independent variables.

Dependent Variable: Crop Yield

Independent Variables: The Fertilizer blend (X, Y, and Z) and type of crop (wheat, corn, soy, and rice).

A two-way ANOVA was used to analyze the data. The dataset includes 60 observations in total, with 5 samples for each combination of fertilizer and crop (3 blends x 4 crops). The null hypothesis states that there is no interaction between fertilizer blend and the type of crop, meaning the fertilizers would have the same effects across all crops.

Descriptive Statistics

The data were summarized by calculating the mean and standard deviation for each group before performing the ANOVA. This gives an initial idea of how the fertilizer blends across different crops.

Based on the results, **Blend Z** generally produces the highest yields compared to the other crops. This was noticeable in soybeans and rice, where you can see the differences. On the other hand, **Blend Y** often showed lower yields, particularly seen in soy and wheat. **Blend X** had moderate results, sometimes close to Blend Z, but not consistently the highest.

It has been observed that the differences between fertilizer blends were not the same for all crops. This could be seen in soy that showed a large gap in yield between Blend Y and the other blends. While the wheat and corn showed smaller differences overall. This means that one fertilizer may work well for one crop but not as well for another.

Two-Way ANOVA Results

The results of the dataset showed that it did not violate the assumptions of normality and equal variances, so it could proceed. There were significant effects of fertilizer, $F(2, 48) = 9.93, p < 0.001$. While the effect of crop type was not significant, $F(3, 48) = 2.57, p = 0.065$. Also there was a significant interaction between fertilizer and crop type $F(6, 48) = 2.35, p = 0.046$.

The significant effect of fertilizer means that the type of fertilizer used does not affect crop yield. However, the crop type alone did not show a significant difference in yield. Since it is significant, it means that the effectiveness of a fertilizer depends on the type of crop. It cannot simply say one fertilizer is the best for all types of crops.

Analysis

For soy, there was a significant difference between fertilizers. Both Blend X and Blend Z produced higher yields compared to Blend Y. This suggests that Blend Y is not a good option for soy.

For rice, Blend Z performed significantly better than Blend X. The difference between the other blends was not enough to be considered significant, but Blend Z still had the highest average yield.

For wheat and corn, there were no significant differences found between fertilizer blends. This means that for these crops, the type of fertilizer may not be significant compared to soy and rice.

Conclusion

In conclusion, the results show that the fertilizer blend has a significant effect on crop yield, but this effect is not the same across all crops. Blend Z generally has the highest yields, especially for soy and rice, while Blend Y tends to perform poorly.

The significant interaction between fertilizer and crop type is a necessary finding. It shows that there is no “best” fertilizer for all crops. Instead, the choice of fertilizer should depend on the specific crop being grown.

Overall, this study highlights the importance of selecting the right fertilizer for each crop in order to achieve better yield results. The farmers and agricultural planners should consider these differences when making decisions about fertilizer use.

Github:

```

import pandas as pd
import scipy.stats as stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

data = []
blends = {
    "X": [
        [123,128,166,151],
        [156,150,178,125],
        [112,174,187,117],
        [100,116,153,155],
        [168,109,195,158]
    ],
    "Y": [
        [135,175,140,167],
        [130,132,145,183],
        [176,120,159,142],
        [120,187,131,167],
        [155,184,126,168]
    ],
    "Z": [
        [156,186,185,175],
        [180,138,206,173],
        [147,178,188,154],
        [146,176,165,191],
        [193,190,188,169]
    ]
}
crops = ["Wheat", "Corn", "Soy", "Rice"]
for blend, rows in blends.items():
    for row in rows:
        for i in range(4):
            data.append([blend, crops[i], row[i]])
df = pd.DataFrame(data, columns=["Fertilizer", "Crop", "Yield"])

print("\n--- DESCRIPTIVE STATISTICS ---")
print(df.groupby(["Fertilizer", "Crop"])["Yield"].agg(["mean", "std"]))

model = ols('Yield ~ C(Fertilizer) * C(Crop)', data=df).fit()

```

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anova_table = sm.stats.anova_lm(model, typ=2)
print("\n--- ANOVA TABLE ---")
print(anova_table)

print("\n--- ASSUMPTION ---")

shapiro = stats.shapiro(model.resid)
print(f"Shapiro-Wilk: W={shapiro.statistic:.3f}, p={shapiro.pvalue:.3f}")

groups = [group["Yield"].values for name, group in
df.groupby(["Fertilizer", "Crop"])]
levene = stats.levene(*groups)
print(f"Levene's Test: W={levene.statistic:.3f}, p={levene.pvalue:.3f}")

anova_table["eta_sq"] = anova_table["sum_sq"] / sum(anova_table["sum_sq"])
print("\n--- EFFECT SIZES ---")
print(anova_table)

f_fert = anova_table.loc["C(Fertilizer)", "F"]
p_fert = anova_table.loc["C(Fertilizer)", "PR(>F)"]
f_crop = anova_table.loc["C(Crop)", "F"]
p_crop = anova_table.loc["C(Crop)", "PR(>F)"]
f_int = anova_table.loc["C(Fertilizer):C(Crop)", "F"]
p_int = anova_table.loc["C(Fertilizer):C(Crop)", "PR(>F)"]
eta_fert = anova_table.loc["C(Fertilizer)", "eta_sq"]
eta_crop = anova_table.loc["C(Crop)", "eta_sq"]
eta_int = anova_table.loc["C(Fertilizer):C(Crop)", "eta_sq"]
print("\n--- APA REPORT ---\n")
print(f"""
A two-way ANOVA was conducted to examine the effects of fertilizer blend
(X, Y, Z) and crop type (Wheat, Corn, Soy, Rice) on crop yield.
There was a significant main effect of fertilizer,
 $F(2, 48) = \{f\_fert:.2f\}$ ,  $p = \{p\_fert:.3f\}$ ,  $\eta^2 = \{eta\_fert:.2f\}$ .
The main effect of crop type was
{'significant' if p_crop < 0.05 else 'not significant'},
 $F(3, 48) = \{f\_crop:.2f\}$ ,  $p = \{p\_crop:.3f\}$ ,  $\eta^2 = \{eta\_crop:.2f\}$ .
There was a significant interaction between fertilizer and crop type,
 $F(6, 48) = \{f\_int:.2f\}$ ,  $p = \{p\_int:.3f\}$ ,  $\eta^2 = \{eta\_int:.2f\}$ .
These results indicate that the effectiveness of fertilizer depends on the
type of crop.""")

```

--- DESCRIPTIVE STATISTICS ---

		mean	std
Fertilizer	Corn	135.4	26.604511
	Rice	141.2	18.820202
	Soy	175.8	16.694310
	Wheat	131.8	29.055120
Y	Corn	159.6	31.277788
	Rice	165.4	14.741099
	Soy	140.2	12.872451
	Wheat	143.2	22.331592
Z	Corn	173.6	20.707487
	Rice	172.4	13.258959
	Soy	186.4	14.570518
	Wheat	164.4	21.054691

--- ANOVA TABLE ---

	sum_sq	df	F	PR(>F)
C(Fertilizer)	8782.90	2.0	9.933347	0.000245
C(Crop)	3411.65	3.0	2.572355	0.064944
C(Fertilizer):C(Crop)	6225.90	6.0	2.347138	0.045555
Residual	21220.40	48.0	NaN	NaN

--- ASSUMPTION ---

Shapiro-Wilk: W=0.980, p=0.440
Levene's Test: W=0.675, p=0.755

--- EFFECT SIZES ---

	sum_sq	df	F	PR(>F)	eta_sq
C(Fertilizer)	8782.90	2.0	9.933347	0.000245	0.221562
C(Crop)	3411.65	3.0	2.572355	0.064944	0.086064
C(Fertilizer):C(Crop)	6225.90	6.0	2.347138	0.045555	0.157058
Residual	21220.40	48.0	NaN	NaN	0.535316

--- APA REPORT ---

A two-way ANOVA was conducted to examine the effects of fertilizer blend (X, Y, Z) and crop type (Wheat, Corn, Soy, Rice) on crop yield. There was a significant main effect of fertilizer, $F(2, 48) = 9.93$, $p = 0.000$, $\eta^2 = 0.22$. The main effect of crop type was not significant, $F(3, 48) = 2.57$, $p = 0.065$, $\eta^2 = 0.09$. There was a significant interaction between fertilizer and crop type, $F(6, 48) = 2.35$, $p = 0.046$, $\eta^2 = 0.16$. These results indicate that the effectiveness of fertilizer depends on the type of crop.